

Prathmesh Thakare

E-Mail: prathmeshthakare501@gmail.com | +44 7767870860 | [LinkedIn](#) | [GitHub](#)

Professional Summary

MSc AI graduate with 3 years of experience in production monitoring within the banking sector. Strong academic foundation in anomaly detection and image classification, supported by hands-on research projects using deep learning and statistical methods in Python. Experienced in root cause analysis, performance analytics, and collaborating with cross-functional teams to support business-critical systems. Seeking an opportunity to apply research and analytical skills to real-world ML problems.

Skills

- **AI & Machine Learning:** Anomaly Detection, Time-Series Analysis, Autoencoders, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, MLOps fundamentals
- **Programming:** Python, JavaScript, MySQL
- **Monitoring & Testing:** Dynatrace (APM, OneAgent, Root Cause Analysis), log Analysis, Incident Management
- **Tools:** JIRA, ServiceNow, BMC Remedy
- **Collaboration & Soft Skills:** Cross-functional Collaboration, Client Communication, Mentoring, Analytical Thinking, Problem Solving.

Key Projects:

Time Series Anomaly Detection

03/2026 – Present

- **Problem Statement:** To study time-series anomaly detection techniques for identifying abnormal patterns in sequential data, such as network traffic, building on prior research experience in anomaly detection.
- **Techniques Used:** Researching statistical and deep learning approaches, including moving averages, ARIMA, LSTM-based autoencoders, and reconstruction-error methods, with a focus on data preprocessing, windowing, and model evaluation for sequential anomaly detection.

IndoFashion Image Classification

08/2025 – 12/2025

- **Problem Statement:** Developed a multi-class image classification solution from the IndoFashion dataset, addressing challenges of limited per-class training data and class similarity.
- **Techniques Used:** Implemented a transfer learning approach using TensorFlow with MobileNetV2, combined with stratified sampling, custom data loading, augmentation, and a two-phase training strategy (feature extraction + fine-tuning).
- **Results Achieved:** Achieved ~78% test accuracy, outperforming ResNet50 (61%), with faster training time and better generalisation across most classes.

Anomaly Detection using GANs

04/2024 – 10/2024

- **Problem Statement:** Evaluated the effectiveness of GANs for detecting anomalies in network traffic data, on the CICIDS 2017 dataset (~3M network flow records) using PyTorch.
- **Techniques Used:** Trained a GAN model with data preprocessing, class balancing, and dimensionality reduction using PCA. Applied stratified sampling and cross-validation for model evaluation, and benchmarked its performance against a CNN model for comparison.
- **Results Achieved:** GANs faced stability and convergence challenges on highly imbalanced and high-dimensional data, with limitations related to training time and model collapse compared to CNNs, highlighting the importance of selecting model architectures based on data characteristics and practical deployment constraints.

Work Experience

Quality Kiosk Technologies Pvt Ltd | Software Engineer

12/2021 – 08/2023

- Worked on Application Performance Management (APM) using Dynatrace Managed for Axis Bank and IndusInd Bank.
- Reduced transaction failures by 30% through proactive monitoring of applications such as Net Banking, Mobile Banking, IMPS/UPI, Savings/FD, and Core Banking.
- Performed Root Cause Analysis (RCA) for live production issues, collaborating with application and synthetic monitoring teams to minimise user impact.
- Delivered performance insights and dashboards to senior technology and business stakeholders to support strategic decision-making.
- Recognised with the "Stellar Award" for identifying and resolving a critical issue in a high-pressure environment.

Ruparel Realty India Pvt Ltd | Associate – IT

11/2020 – 12/2021

- Handled BMC Remedy tickets for issue resolution and request tracking, ensuring timely closure and SLA compliance.
- Managed incidents and RCA reporting, and coordinated weekly meetings with Service Delivery Management teams.

Documented issue resolutions and streamlined support workflows, improving internal team efficiency.

Education:

Aston University, Birmingham (2025)

M.Sc. Artificial Intelligence with Business Strategy

Key Modules: Machine Learning, Deep Learning, Mathematics for AI, Leadership and Decision Making, Strategic Management.

Mumbai University, Mumbai (2020)

B.Sc. Information Technology

Programming Language – C, C++, Java, JavaScript, Python, DBMS, MySQL, Linux.